



Presentation 3: Time-series Trends

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CAR Data Drill Down - Weekly Trends

Insights:

- Monday has the highest volume of calls on avg. for all lines except immigration, which gradually increases w/ a peak on Wednesday and then declines
- Looking at avg. calls by 3rd activity in Family line:
 - 5–6 more callers that end up abandoning the call instead of going to the Simple Divorce line
 - And about 12 more callers on avg. that abandon instead of going to Consumer queue
 - Process for getting into Consumer/SD queue may be confusing or long, so callers hang up
- Consumer queue is very underutilized in the Housing line as well, with about 1–3 callers in Next activity and 3rd activity
- Avg. # of abandoned calls is much higher in Housing than Family in Next Activity (around 35 calls compared to 1)
 - Callers are not going to the Consumer Queue or LegalMenu2

Recommendations :

- Make prompts for lines as concise as possible using AI tools/LLM
- Create a note on the website with optimal day of the week to call back for particular EP line

CAR data Call Flows: Main Menu Traffic

Jasmine

1. Seasonal trends starting from LAC's Main Menu
 - a. Spike in callers with legal issues in January, another rise in summer/fall
 - i. Trend for both seniors and non-seniors
 - b. Spike in non-seniors seeking legal help family-related matters in January, rise in the summer/fall driven by family and housing matters
 - c. For seniors, spike in housing matters earlier in the year, increased callers seeking help with benefits in latter half of the year.

[Dashboard link](#)



CAR data Call Flows: Main Menu Traffic

Jasmine

2. Weekly trends starting from LAC's Main Menu

- a. Highest call volume in the beginning of the week, particularly for legal issues
 - i. Trend most apparent for non-seniors; seniors' call volume is more dispersed throughout the week
- b. For non-seniors, consistently highest call volume for family and housing menus throughout the week
 - i. Family menu sees the most significant weekly trend
- c. Seniors are most likely to encounter a closed queue on Fridays vs. other days of the week
 - i. May reflect staffing, call patterns of other calls, etc.

→
[Dashboard link](#)

CAR data Call Flows: Main Menu Traffic

Jasmine

3. Hourly trends starting from LAC's Main Menu

- a. Highest call volume early in the day, driven by callers for legal issues
 - i. Trend most apparent for non-seniors; seniors' call volume is more dispersed throughout the day
- b. For non-seniors, most significant trend in call volume for family and housing menus throughout the day

[Dashboard link](#)



Called Number Analysis

Business Relevance: by looking at the call volume of inbound calls to each number, we can determine if we should increase the staffing for certain lines or merge multiple lines

Method: filtered 300+ unique called numbers and kept only 20+ numbers for which we have a clear understanding of their services

Improvements since last version:

- Replace the exact numbers with their descriptions
- Standardized call volume by calculating the average calls per weekday
- Mapped call flows by tracking each leg of a call, visualizing how call volume changes even for transferred numbers

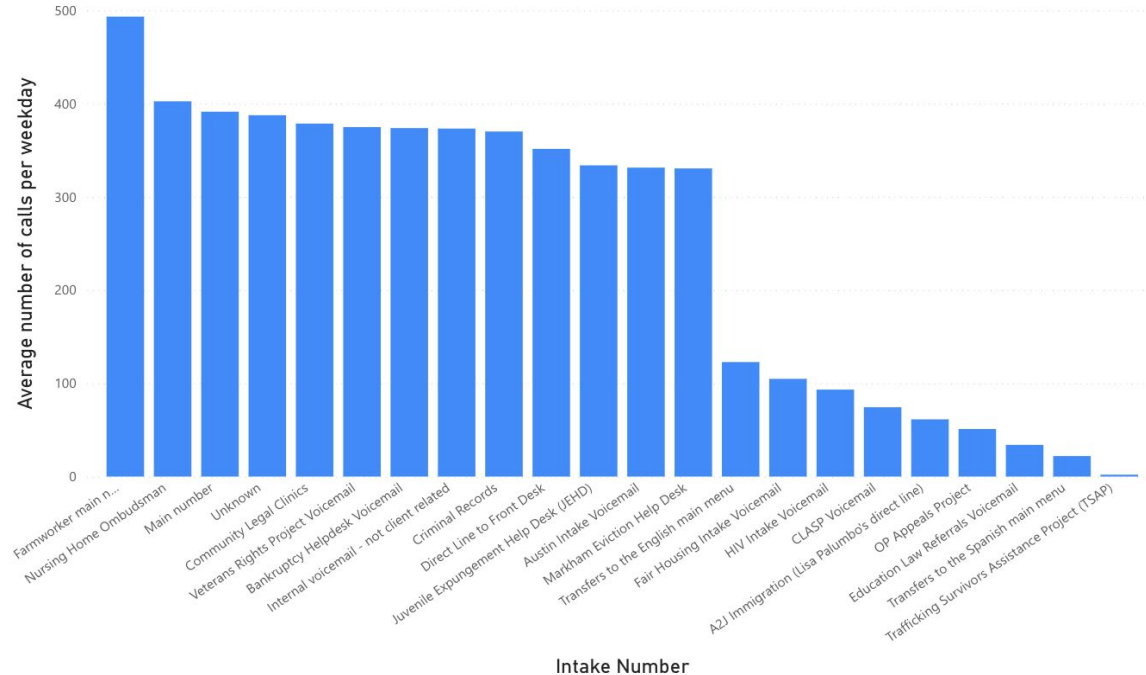
Map out the call flows

- Sort by Correlation ID and Start Time to ensure proper order
- Group Correlation ID and collect all number descriptions from each leg [**'Call Flow'**]
- Count the number of “legs” (steps) in each call [**'num_legs'**]
- Expand each call leg horizontally—each column represents a leg of the call.

	Correlation ID	Call Flow	num_legs	Number Description_1	Number Description_2	Number Description_3	Number Description_4
0	00000fe9-dfa0-41c5-986b-d9ce731f2715	[Main number]	1	Main number	None	None	None
1	00001e73-ce33-48f1-b531-bddc7ee3965d	[Farmworker main number / Migrant Legal Assist...]	1	Farmworker main number / Migrant Legal Assista...	None	None	None
2	00006ccc-8250-4993-90c0-bbfd41f7dd24	[Unknown, Internal voicemail - not client rela...]	3	Unknown	Internal voicemail - not client related	Internal voicemail - not client related	None
3	00008bec-c404-48cc-8276-1f276bf4229a	[Bankruptcy Helpdesk Voicemail, Internal voice...]	3	Bankruptcy Helpdesk Voicemail	Internal voicemail - not client related	Internal voicemail - not client related	None
4	0000aef0-6f35-4560-a4be-5b3bc31fbefb	[Main number]	1	Main number	None	None	None
5	0000ff52-cebd-430b-bca8-b8ae9a8b6e04	[Main number]	1	Main number	None	None	None
6	0001424d-5229-46a5-9661-04dbc11e5273	[Unknown, Internal voicemail - not client rela...]	3	Unknown	Internal voicemail - not client related	Internal voicemail - not client related	None
8	000197f8-fafe-46d3-a830-94aefc90fd90	[Internal voicemail - not client related]	1	Internal voicemail - not client related	None	None	None
9	0001af3f-5951-4d02-bf61-27cab9793b9d	[Unknown, Internal voicemail - not client rela...]	3	Unknown	Internal voicemail - not client related	Internal voicemail - not client related	None
10	0001bba9-cd7c-4e4d-94be-c54e880c10c5	[Main number]	1	Main number	None	None	None

Call volume for the intake numbers (average per weekday)

Call volume for the intake numbers

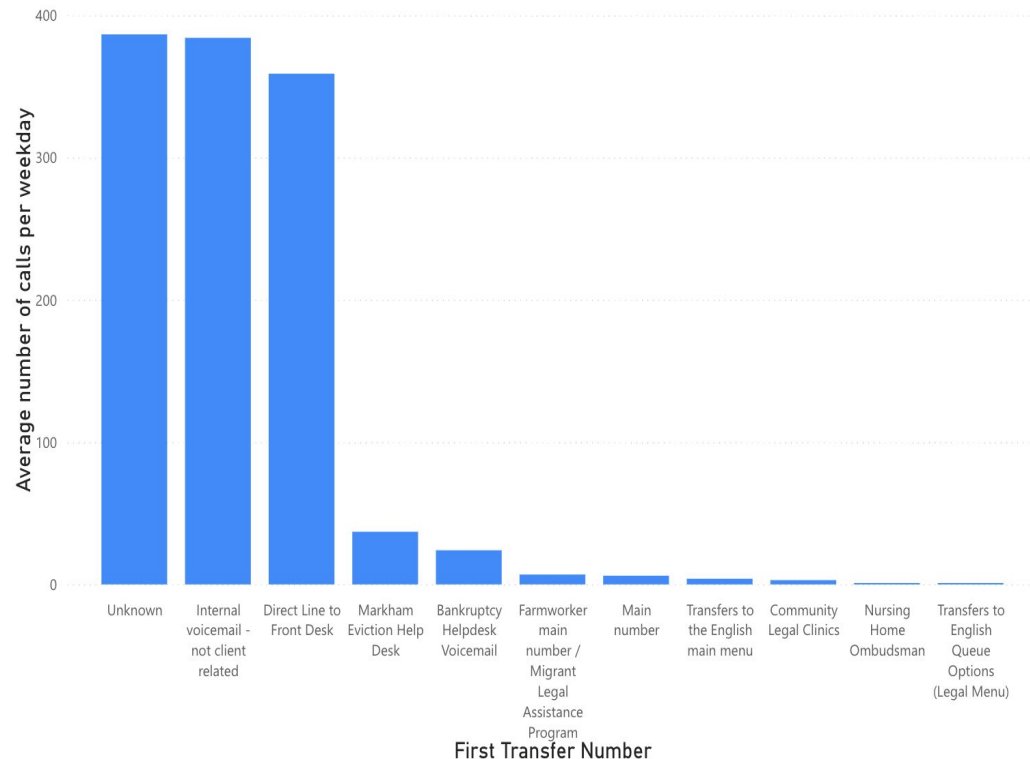


Findings:

- **Farmworker main number** received the most number of calls, followed by **Nursing Home Ombudsman** and **Main number**
- While the average number of call per weekday is relatively the high from Farmworker main number to Markham Eviction Help Desk, there's a huge drop from Transfer to English main menu
- Low call volume for A2J immigration, OP Appeals Project, Trafficking Survivors Assistance Project

Call volume for first transfer numbers (average per weekday)

Call volume for the first transfer numbers



Intake number

Number	Description_1
A2J Immigration (Lisa Palumbo's direct line)	Internal voicemail - not client related
Austin Intake Voicemail	Juvenile Expungement Help Desk (JEHD)
Bankruptcy Helpdesk Voicemail	Main number
CLASP Voicemail	Markham Eviction Help Desk
Community Legal Clinics	Nursing Home Ombudsman
Criminal Records	OP Appeals Project
Direct Line to Front Desk	Trafficking Survivors Assistance Project
Education Law Referrals Voicemail	Transfers to the English main menu
Fair Housing Intake Voicemail	Transfers to the Spanish main menu
Farmworker main number / Migrant Legal Assistance Program	Unknown
HIV Intake Voicemail	Veterans Rights Project Voicemail

Findings:

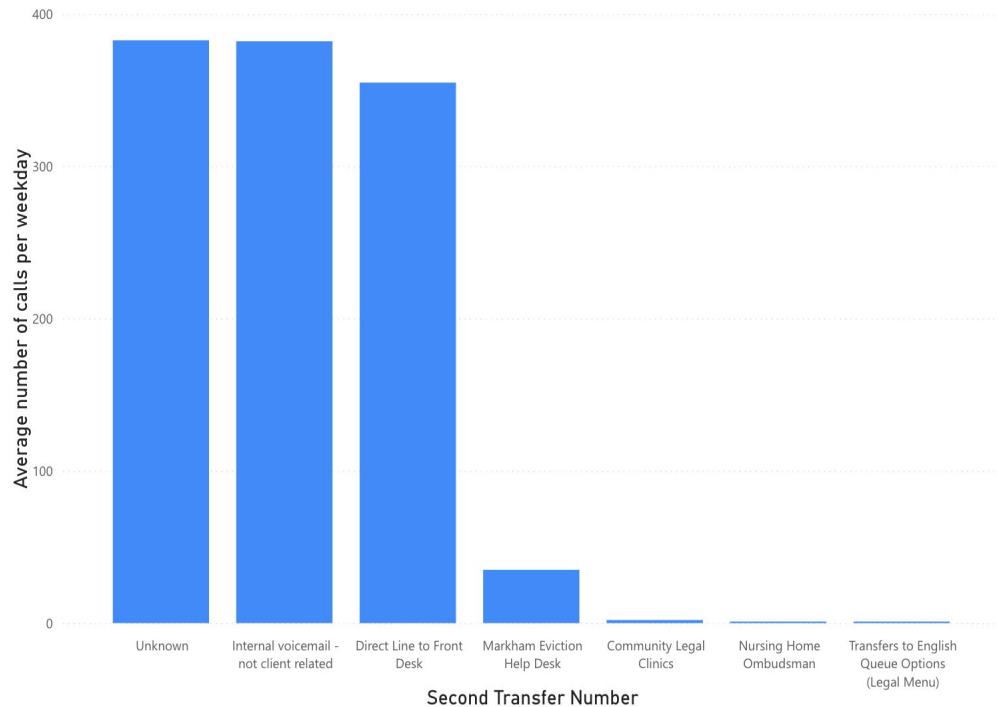
- Most calls are transferred to **unknown numbers** without a description of their services
- The remaining calls are transferred to internal voicemail or directly to the front desk

Next Step:

- The high volume of transfers to unknown numbers highlights the need to understand their purposes, which would improve insight into overall call flows.

Call volume for second transfer numbers (average per weekday)

Call volume for the second transfer number



Intake number
Number Description_1

A2J Immigration (Lisa Palumbo's direct line)	Internal voicemail - not client related
Austin Intake Voicemail	Juvenile Expungement Help Desk (JEHD)
Bankruptcy Helpdesk Voicemail	Main number
CLASP Voicemail	Markham Eviction Help Desk
Community Legal Clinics	Nursing Home Ombudsman
Criminal Records	OP Appeals Project
Direct Line to Front Desk	Trafficking Survivors Assistance Project
Education Law Referrals Voicemail	Transfers to the English main menu
Fair Housing Intake Voicemail	Transfers to the Spanish main menu
Farmworker main number / Migrant Legal Assistance ...	Unknown
HIV Intake Voicemail	Veterans Rights Project Voicemail

Findings:

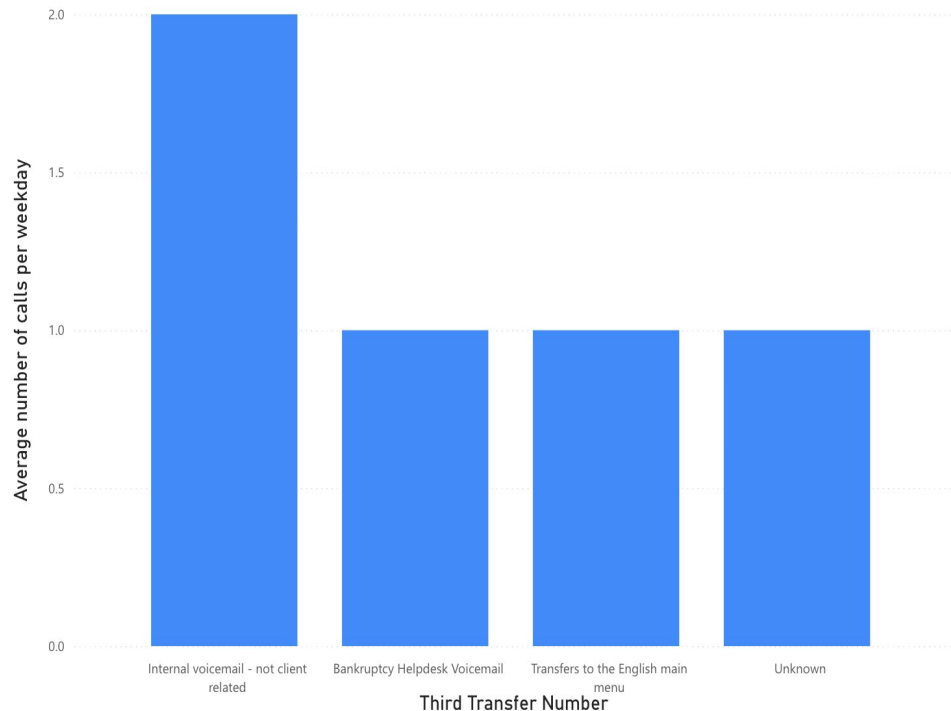
- Unknown, Internal Voicemail and Direct Line to Front Desk still consist of the most second numbers

Next Step:

- Based on the description, internal voicemail is not client related, but it's still one of the most popular transfer numbers for inbound calls in the All Calls Data. We should understand how this number is used for inbound calls.

Call volume for third transfer numbers (average per weekday)

Call volume for the third transfer number



Intake number

Number Description_1

A2J Immigration (Lisa Palumbo's direct line)	Internal voicemail - not client related
Austin Intake Voicemail	Juvenile Expungement Help Desk (JEHD)
Bankruptcy Helpdesk Voicemail	Main number
CLASP Voicemail	Markham Eviction Help Desk
Community Legal Clinics	Nursing Home Ombudsman
Criminal Records	OP Appeals Project
Direct Line to Front Desk	Trafficking Survivors Assistance Projec...
Education Law Referrals Voicemail	Transfers to the English main menu
Fair Housing Intake Voicemail	Transfers to the Spanish main menu
Farmworker main number / Migrant Legal Assistance ...	Unknown
HIV Intake Voicemail	Veterans Rights Project Voicemail

Findings:

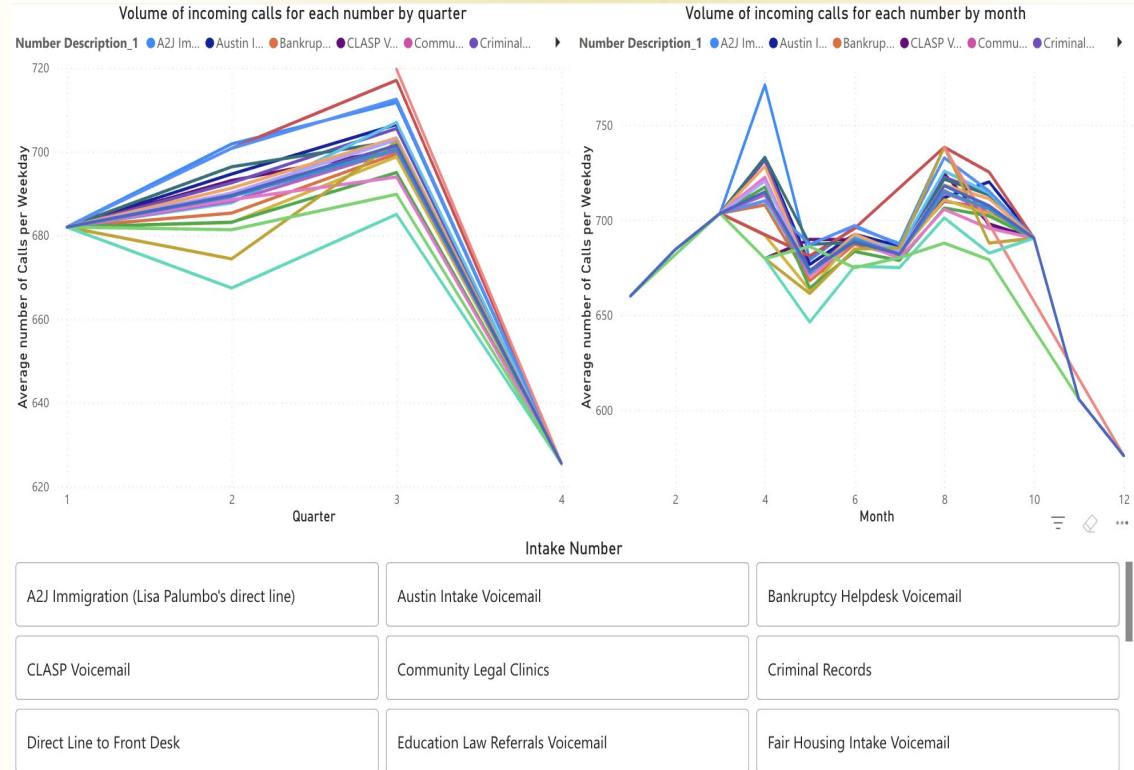
- Only consist of Internal Voicemail, Bankruptcy Helpdesk Voicemail, Transfer to the English Main menu, and Unknown

Next Step:

- Investigate the previous transfer numbers to see how to improve transfer efficiency

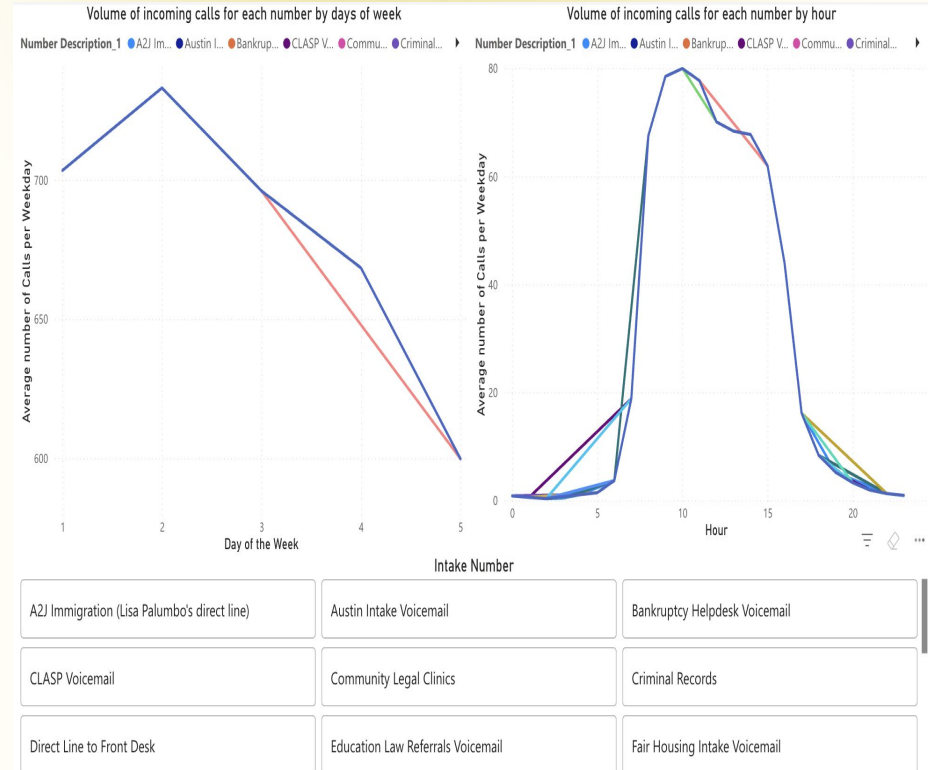
Volume of Incoming Calls by Quarter and Month

- There is an average of 700 calls made per weekday, and this trend is seen across months and quarters.
- Immigration peaks the highest of all numbers in the month of April, and again in August, however in quarterly trends it peaks during the start quarter three.
- The Farmworker and HIV line are consistently called the least compared to other lines both monthly and quarterly.



Volume of Incoming Calls by Weekday and Hour

- The standardized plots for the incoming calls by hour and days of the week show that there is a normal distribution for the calls during the day, with the peak around 10 am at about 80 calls.
- In contrast, across days of the week, there is a steep decline as the week continues, with Tuesday having the most calls at an average of 733 calls per day.
- Friday typically has the lowest number of calls at about 600 calls per day.



Next Steps

Further Investigation:

- More analysis is needed of the volume of incoming calls per week because many of the different numbers have overlapping distributions, which seem too similar.
- Further analysis into the relationship between monthly and quarterly trends.